

AERO40006 - Mathematics 1

Tutorial 3

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Question 1

Determine if the following series converge:

(1)	$\sum_{n \geq 0} \frac{(n+1)^2}{2^n}$	(2)	$\sum_{n \geq 1} e^{-n}$	(3)	$\sum_{n \geq 1} \frac{n^{100}}{2^n + 5}$
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$$c1) \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+2)^2}{2^{n+1}}}{\frac{(n+1)^2}{2^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+2)^2}{2^{n+1}} \cdot \frac{2^n}{(n+1)^2} \right| = \lim_{n \rightarrow \infty} \left(\frac{1}{2} \cdot \frac{(n+2)^2}{(n+1)^2} \right) < 1 \quad \boxed{CV}$$

$$c2) \lim_{n \rightarrow \infty} \left| \frac{e^{-(n+1)}}{e^{-n}} \right| = \lim_{n \rightarrow \infty} \left| e^{-n-1} \cdot e^n \right| = \lim_{n \rightarrow \infty} \left| e^{-1} \right| = e^{-1} < 1 \quad \boxed{CV}$$

$$c3) \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)^{100}}{2^{n+1} + 5}}{\frac{n^{100}}{2^n + 5}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^{100}}{2^{n+1} + 5} \cdot \frac{2^n + 5}{n^{100}} \right| = \lim_{n \rightarrow \infty} \left| \left(\frac{n+1}{n} \right)^{100} \cdot \frac{2^n + 5}{2^{n+1} + 5} \right|$$

$$= 1^{100} \cdot \frac{1}{2}$$

$$= \frac{1}{2} < 1 \quad \boxed{CV}$$

Question 2

Determine if the following series converge:

(1) $\sum_{n=0}^{\infty} nx^n$	(2) $\sum_{n=0}^{\infty} \frac{n^2}{2^n} (x-1)^n$	(3) $\sum_{n \geq 1} \frac{(n+2)!}{(2n)!} x^n$
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$$(1) \quad \lim_{n \rightarrow \infty} \frac{|(n+1)x^{n+1}|}{|nx^n|} = \lim_{n \rightarrow \infty} \left| \frac{(n+1)x^{n+1}}{nx^n} \right| = |x|$$

If $x < 1$ and $x > -1$, the series converge

If $x = 1$, $\sum_{n=0}^{\infty} nx^n = \sum_{n=0}^{\infty} n$, the series diverge

If $x = -1$, $\sum_{n=0}^{\infty} nx^n = \sum_{n=0}^{\infty} (-1)^n n$, $\lim_{n \rightarrow \infty} n \neq 0$, the series diverge

$\therefore$ The series converge when $-1 < x < 1$

$$(2) \quad \lim_{n \rightarrow \infty} \frac{\left| \frac{(n+1)^2 (x-1)^{n+1}}{2^{n+1}} \right|}{\left| \frac{n^2 (x-1)^n}{2^n} \right|} = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2 (x-1)^{n+1}}{2^{n+1}} \cdot \frac{2^n}{n^2 (x-1)^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \left(\frac{n+1}{n} \right)^2 \frac{x-1}{2} \right| = \frac{1}{2} |x-1|$$

If $x < 3$ and $x > -1$, the series converge

If $x = 3$, $\sum_{n=0}^{\infty} \frac{n^2}{2^n} (x-1)^n = \sum_{n=0}^{\infty} \frac{n^2}{2^n} \cdot 2^n = \sum_{n=0}^{\infty} n^2$, the series diverge

If $x = -1$, $\sum_{n=0}^{\infty} \frac{n^2}{2^n} (x-1)^n = \sum_{n=0}^{\infty} \frac{n^2}{2^n} \cdot (-2)^n = \sum_{n=0}^{\infty} (-1)^n n^2$

$\therefore \lim_{n \rightarrow \infty} n^2 \neq 0 \quad \therefore$ the series diverge

$\therefore$ The series converge when $-1 < x < 3$

Question 2

Determine if the following series converge:

(1) $\sum_{n=0}^{\infty} nx^n$	(2) $\sum_{n=0}^{\infty} \frac{n^2}{2^n} (x-1)^n$	(3) $\sum_{n \geq 1} \frac{(n+2)!}{(2n)!} x^n$
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$$\begin{aligned}
 (3) \quad \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+3)! x^{n+1}}{(2n+1)!}}{\frac{(n+2)! x^n}{(2n)!}} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+3)! x^{n+1}}{(2n+1)!} \cdot \frac{(2n)!}{(n+2)! x^n} \right| \\
 &= \lim_{n \rightarrow \infty} \left| (n+3) \cdot x \cdot \frac{1}{2n+1} \right| \\
 &= \lim_{n \rightarrow \infty} \left| \frac{n+3}{2n+1} x \right| \\
 &= \frac{1}{2} |x|
 \end{aligned}$$

If $x < 2$ and $x > -2$, the series converge

If $x = 2$, $\sum_{n \geq 1} \frac{(n+2)!}{(2n)!} 2^n$, the series converge

If $x = -2$, $\sum_{n \geq 1} \frac{2^n (n+2)!}{(2n)!} (-1)^n$, $\lim_{n \rightarrow \infty} \frac{(n+2)!}{(2n)!} 2^n = 0$

$\therefore$ the series converge *for any x*

$\therefore$ The series converge when $-2 < x < 2$

Question 3

Determine if the following series converge:

(1)	$\sum_{n=1}^{\infty} \frac{1}{n}$	(2)	$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$	(3)	$\sum_{n=1}^{\infty} \frac{1}{n^2}$
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$$\langle 1 \rangle \quad \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x} dx = \lim_{n \rightarrow \infty} [\ln|x|]_1^n = \lim_{n \rightarrow \infty} \ln n = \infty \quad \boxed{\text{DV}}$$

$$\langle 2 \rangle \quad \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^{\frac{1}{2}}} dx = \lim_{n \rightarrow \infty} [2x^{\frac{1}{2}}]_1^n = \lim_{n \rightarrow \infty} (2\sqrt{n} - 2) = \infty \quad \boxed{\text{DV}}$$

$$\langle 3 \rangle \quad \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^2} dx = \lim_{n \rightarrow \infty} [-x^{-1}]_1^n = \lim_{n \rightarrow \infty} (1 - \frac{1}{n}) = 1 < \infty \quad \boxed{\text{CV}}$$

Question 4

Expand the function

$$f(x) = \frac{1}{1-x}$$

around $x = 0$ in a series up to the first four non-zero terms. Use this expansion to find an approximation for $f(x)$ at $x = 0.1$. Using the remainder term, what error bound can you give for the difference between the exact value of $f(x)$ at $x = 0.1$ and the approximation computed from the series expansion?

$$f(x) = (1-x)^{-1}$$

$$f'(x) = (1-x)^{-2}$$

$$f''(x) = 2(1-x)^{-3}$$

$$f'''(x) = 6(1-x)^{-4}$$

$$f^{(4)}(x) = 24(1-x)^{-5}$$

About $x = 0$,

$$f(x) = 1 + x + x^2 + x^3$$

$$\therefore f(0.1) = 1 + (0.1) + (0.1)^2 + (0.1)^3 = 1.111$$

$$R_n = \frac{f^{(4)}(x_0)}{4!} x^4 = \frac{\frac{x^4}{(1-x_0)^5}}{x^4(x_0)} = (0.1)^4(x_0)$$

$$\therefore 0 \leq R_n \leq 1 \times 10^{-5} \quad 1.7 \times 10^{-4}$$

$$\therefore 1.111 \leq f(0.1) \leq 1.11101 \quad f(0.1) - 1.111 \leq 1.7 \times 10^{-4}$$

Question 5

A publishing company gives an academic a profit of \$100 for the first year for a book (yes, this is not a lot!). For each subsequent year the profit falls by 9%. Determine the total profit possible.

$$\begin{aligned}
 \text{Total profit (\$)} &= TP \\
 &= 100 + 100 \times (1 - 9\%) + 100 \times (1 - 9\%)^2 + \dots + 100 \times (1 - 9\%)^n \\
 &= 100 + 100(0.91) + 100(0.91)^2 + \dots + 100(0.91)^n \\
 \therefore (0.91)TP &= 100(0.91) + 100(0.91)^2 + 100(0.91)^3 + \dots + 100(0.91)^{n+1} \\
 \therefore (1 - 0.91)TP &= 100 - 100(0.91)^{n+1} \\
 \therefore TP &= \frac{10000}{9} - \frac{10000}{9}(0.91)^{n+1} \\
 \therefore TP_{n \rightarrow \infty} &= \frac{10000}{9} \\
 &= 1111.11 (\$)
 \end{aligned}$$

Question 6

Find the 5th degree Taylor polynomial for the function $f(x) = \cos(x)$ at $a = -\frac{\pi}{2}$.

$$f(x) = \cos(x)$$

$$f'(x) = -\sin(x)$$

$$f''(x) = -\cos(x)$$

$$f'''(x) = \sin(x)$$

$$f^{(4)}(x) = \cos(x)$$

$$f^{(5)}(x) = -\sin(x)$$

$$\begin{aligned}\therefore f(x) &= 0 + 1 \cdot \left(x + \frac{\pi}{2}\right) + 0 + \frac{(-1)}{3!} \left(x + \frac{\pi}{2}\right)^3 + 0 + \frac{1}{5!} \left(x + \frac{\pi}{2}\right)^5 \\ &= x + \frac{\pi}{2} - \frac{1}{6} \left(x + \frac{\pi}{2}\right)^3 + \frac{1}{120} \left(x + \frac{\pi}{2}\right)^5\end{aligned}$$

Question 7

For each of the following functions, state whether they are odd, even, or neither odd nor even.

- (a) x^n (c) $x^3 - x$ (e) $\cos(x) \sin(x^2)$
 (b) $7x^4 - 4x^2 + 1$ (d) xe^x (f) $\cosh(x)$

(a) If n is odd, $(-x)^n = -x^n$, x^n is odd

If n is even, $x^n = (-x)^n$, x^n is even

(b) $7(-x)^4 - 4(-x)^2 + 1 = 7x^4 - 4x^2 + 1$, $7x^4 - 4x^2 + 1$ is even

(c) $(-x)^3 - (-x) = -x^3 + x = -(x^3 - x)$, $x^3 - x$ is odd

(d) $(-x)e^{-x} \neq -xe^x \neq xe^x$ neither odd nor even

(e) $\cos(-x) \sin(x^2) \neq -\cos(x) \sin(x^2) \neq \cos(x) \sin(x^2)$
 neither odd nor even

$$(f) \cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\frac{e^{(-x)} + e^x}{2} = \frac{e^x + e^{-x}}{2} \text{ is even}$$

Question 8

Prove that if $f(x)$ is an even function then

$$\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx.$$

$$\int_{-L}^L f(x) dx = \int_0^L f(x) dx + \int_{-L}^0 f(x) dx$$

$$\text{Let } u = -x, \quad \frac{du}{dx} = -1 \quad \therefore dx = -du$$

$$\therefore \int_{-L}^L f(x) dx = \int_0^L f(x) dx + \int_L^0 f(-u) (-du)$$

$$= \int_0^L f(x) dx + \int_0^L f(-u) du$$

$$= \int_0^L f(x) dx + \int_0^L f(u) dx$$

$$= \int_0^L f(x) dx + \int_0^L f(-x) dx$$

$\therefore f(x)$ is an even function

$$\therefore f(-x) = f(x)$$

$$\therefore \int_{-L}^L f(x) dx = \int_0^L f(x) dx + \int_0^L f(x) dx$$

$$= 2 \int_0^L f(x) dx$$

Question 9

Find the Fourier series of

$$f(x) = \begin{cases} \cos x & -\pi < x \leq 0 \\ -\cos x & 0 < x \leq \pi \end{cases}$$

and use Parseval's theorem to show that

$$\sum_{n=1}^{\infty} \frac{n^2}{(2n-1)^2(2n+1)^2} = \frac{\pi^2}{64}.$$

$$a_0 = 0, \quad a_n = 0, \quad b_n = \frac{2}{\pi} \int_0^{\pi} \cos(x) \sin(nx) dx$$

$$\int_0^{\pi} \cos(x) \sin(nx) dx = -\sin(x) \sin(nx) - n \cos(x) \cos(nx) + n^2 \int_0^{\pi} \cos(x) \sin(nx)$$

$$\therefore \int_0^{\pi} \cos(x) \sin(nx) dx = \left(\frac{1}{n^2-1} \right) (-\sin(x) \sin(nx) - n \cos(x) \cos(nx))$$

$$\begin{aligned} \therefore b_n &= \frac{2}{\pi(n^2-1)} [-\sin(x) \sin(nx) - n \cos(x) \cos(nx)]_0^{\pi} = \frac{2}{\pi(n^2-1)} [(0 + n \cos(n\pi)) - (0 - n \cos(n\pi))] \\ &= \frac{4}{\pi} \frac{n}{(n+1)(n-1)} \cos(n\pi) \end{aligned}$$

Let $n = 2n$

$$\therefore b_n = \frac{8}{\pi} \frac{n}{(n+1)(n-1)} \cos(2n\pi) = \frac{8}{\pi} \frac{n}{(n+1)(n-1)}$$

$$\frac{1}{L} \int_{-L}^L f(x)^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

$$\therefore \frac{2}{\pi} \int_0^{\pi} \cos^2(x) dx = \frac{2}{\pi} \int_0^{\pi} \frac{1 + \cos(2x)}{2} dx = \frac{1}{\pi} \int_0^{\pi} (1 + \cos(2x)) dx$$

$$= \frac{1}{\pi} \left[x + \frac{1}{2} \sin(2x) \right]_0^{\pi} = 1$$

$$= \sum_{n=1}^{\infty} b_n^2 = \sum_{n=1}^{\infty} \frac{64}{\pi^2} \frac{n^2}{(n+1)^2(n-1)^2}$$

$$\therefore \sum_{n=1}^{\infty} \frac{n^2}{(n+1)^2(n-1)^2} = \frac{\pi^2}{64}$$

Question 10

At time $t > 0$ and location $0 < x < 2$, the function

$$U(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) e^{-n^2 \pi^2 t}$$

represents the heat in a wire with endpoints kept at zero temperature. Find the constants b_n in the case that the initial temperature distribution is

$$U(x, 0) = 3, \quad 0 < x < 2.$$

$$U(x, 0) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) = 3$$

Make an odd extension for $U(x, 0)$ as $f(x)$ from 0 to 2

$$\therefore f(x) = \begin{cases} -3 & -2 < x < 0 \\ 3 & 0 < x < 2 \end{cases}$$

$$\therefore b_n = \frac{2}{2} \int_0^2 3 \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= 3 \int_0^2 \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{6}{n\pi} \left[-\cos\left(\frac{n\pi x}{2}\right) \right]_0^2$$

$$= \frac{6}{n\pi} (-\cos(n\pi) + 1)$$

$$\therefore b_n = \begin{cases} n \text{ is even, } 0 \\ n \text{ is odd, } \frac{12}{n\pi} \end{cases}$$

Question 11

(i) Find the exponential Fourier series for the square wave function

$$f(x) = \begin{cases} 1 & -1 < x \leq 0 \\ 2 & 0 < x < 1 \end{cases}$$

(ii) What is the value of the resulting Fourier series at $x = 0$?

$$c_n = \frac{1}{2L} \int_{-1}^1 f(x) e^{-\frac{in\pi x}{L}} dx \quad \because L = 1$$

$$\begin{aligned} \therefore C_n &= \frac{1}{2} \int_{-1}^1 f(x) e^{-in\pi x} dx = \frac{1}{2} \left(\int_{-1}^0 1 e^{-in\pi x} dx + \int_0^1 2 e^{-in\pi x} dx \right) \\ &= \frac{1}{2} \left(\frac{1}{-in\pi} [e^{-in\pi x}]_{-1}^0 + \frac{2}{-in\pi} [e^{-in\pi x}]_0^1 \right) \\ &= \frac{1}{2} \left(\frac{1}{-in\pi} + \frac{e^{in\pi}}{in\pi} + \frac{2e^{-in\pi}}{-in\pi} + \frac{2}{in\pi} \right) = \frac{1}{2in\pi} + \frac{(e^{in\pi} - 2e^{-in\pi})}{2in\pi} \\ &= \frac{1}{2in\pi} + \frac{\cos(n\pi) + \sin(n\pi) - 2\cos(n\pi) + 2\sin(n\pi)}{2in\pi} \\ &= \frac{1}{2in\pi} + \frac{3\sin(n\pi) - \cos(n\pi)}{2in\pi} = \frac{1 - \cos(n\pi)}{2in\pi} \\ &= \frac{i}{n\pi} \quad (n \text{ is odd}) \quad \text{OR} \quad 0 \quad (n \text{ is even}) \end{aligned}$$

$$\therefore \frac{1}{2} a_0 = C_0 = \frac{1}{2} \left(\int_{-1}^0 dx + \int_0^1 2 dx \right) = \frac{3}{2}$$

$$\begin{aligned} \therefore F(x) &= \frac{3}{2} - \sum_{n=-\infty}^{\infty} C_n e^{\frac{in\pi x}{L}} \\ &= \frac{3}{2} - \frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{i}{(2n-1)} e^{i(2n-1)\pi x} \end{aligned}$$

$$\text{ii) } C_0 = \frac{3}{2}$$